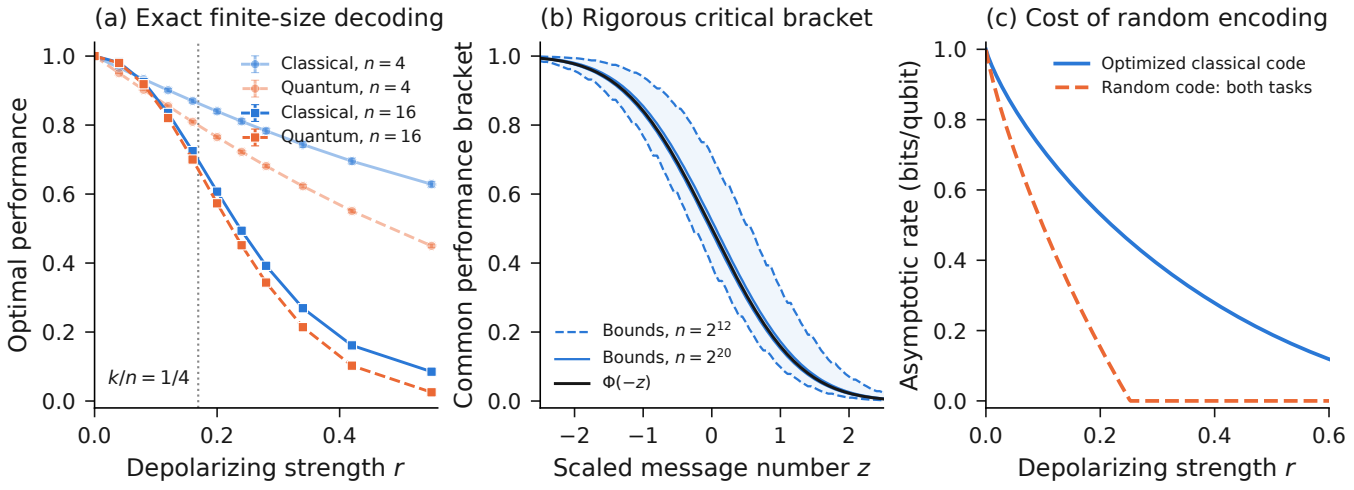
(a) Exact finite-size decoding
Optimal performance
Depolarizing strength r
Classical, $n = 4$
Quantum, $n = 4$
Classical, $n = 16$
Quantum, $n = 16$
$k/n = 1/4$
(b) Rigorous critical bracket
Common performance bracket
Scaled message number z
Bounds, $n = 2^{12}$
Bounds, $n = 2^{20}$
$\Phi(-z)$
(c) Cost of random encoding
Asymptotic rate (bits/qubit)
Depolarizing strength r
Optimized classical code
Random code: both tasks